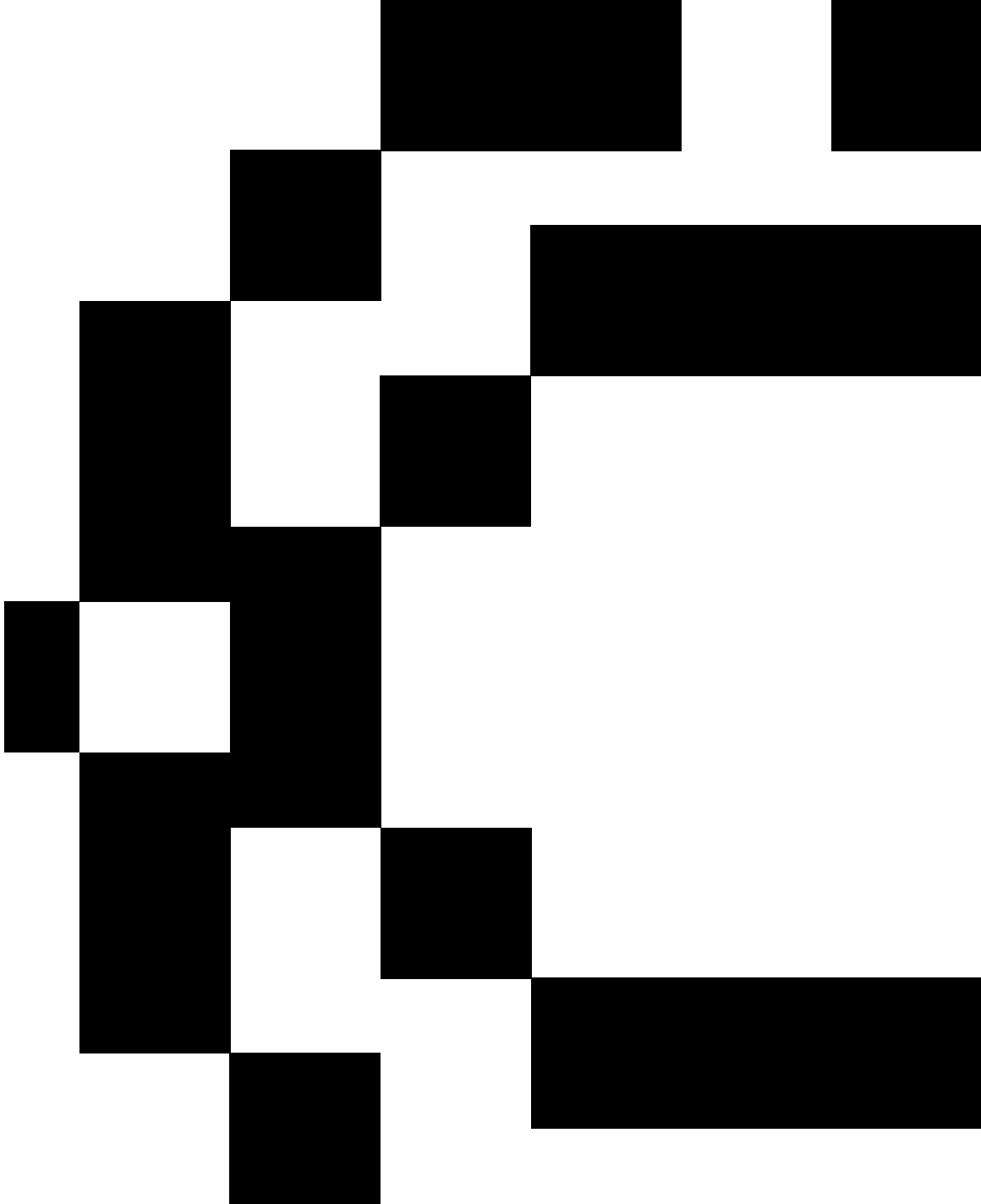
NOVA IMS INFORMATION MANAGEMENT SCHOOL

Computation I / Introduction to Programming

Practicals Week 06 - Introduction to For Loops

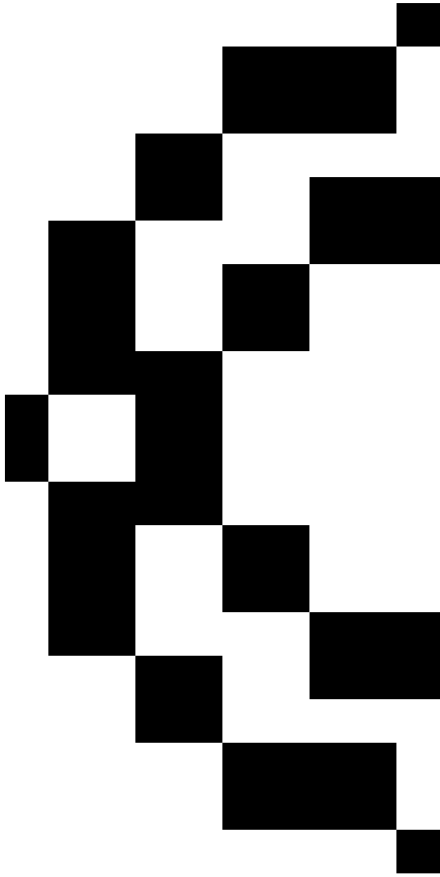


Exercise 1

Using a `for` loop, print all the integers from 1 to 10 inclusive, separated by spaces on a single line.

Example

- `Input:  (no input)`
- `Output: 1 2 3 4 5 6 7 8 9 10`



Exercise 1 — Solution

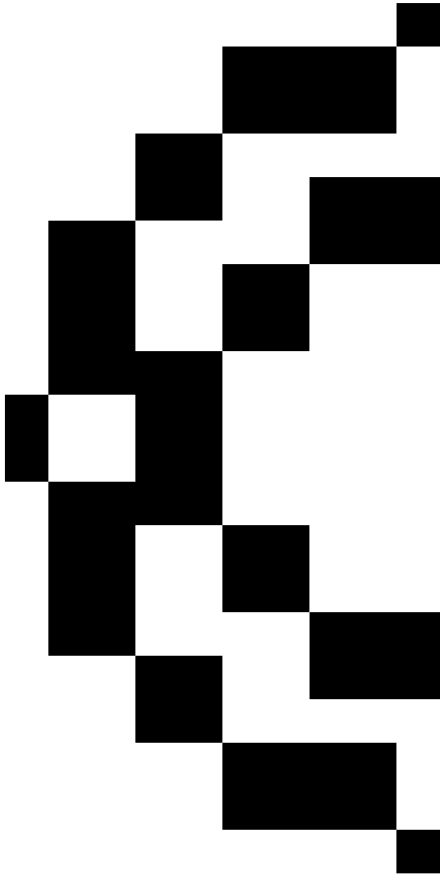
```
for i in range(1, 11):
    print(i, end=" ")
print()
```

Exercise 2

Using a `for` loop, print all the numbers from 1 to 100 that are divisible by 3.

Example

```
Input:  (no input)
Output: 3
        6
        9
        ...
        99
```



Exercise 2 — Solution

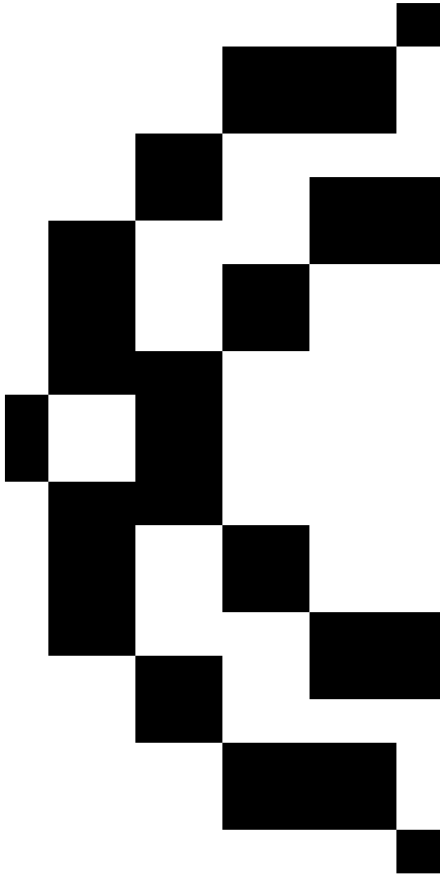
```
for i in range(3, 101, 3):
    print(i)
```

Exercise 3

Ask the user for an integer `n`, then build a list containing all the integers from 1 to `n` inclusive, using a `for` loop and `.append()`.

Example

- `Input:  n = 5`
- `Output: [1, 2, 3, 4, 5]`



Exercise 3 — Solution

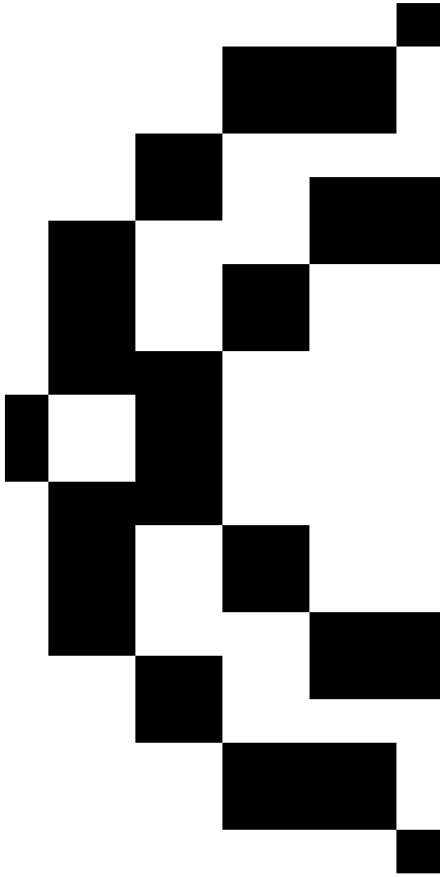
```
n = int(input("Insert a number: "))
my_list = []
for i in range(n):
    my_list.append(i + 1)
print(my_list)
```

Exercise 4

Using a `for` loop, print all the numbers from 1 to 100 on a single line, separated by spaces. Whenever the running count reaches a multiple of 10, print the message `"That's 10 numbers printed!"` on its own line before continuing.

Example

```
Input:  (no input)
Output: 1 2 3 4 5 6 7 8 9 10
        That's 10 numbers printed!
        11 12 13 14 15 16 17 18 19 20
        That's 10 numbers printed!
        ...
```



Exercise 4 — Solution

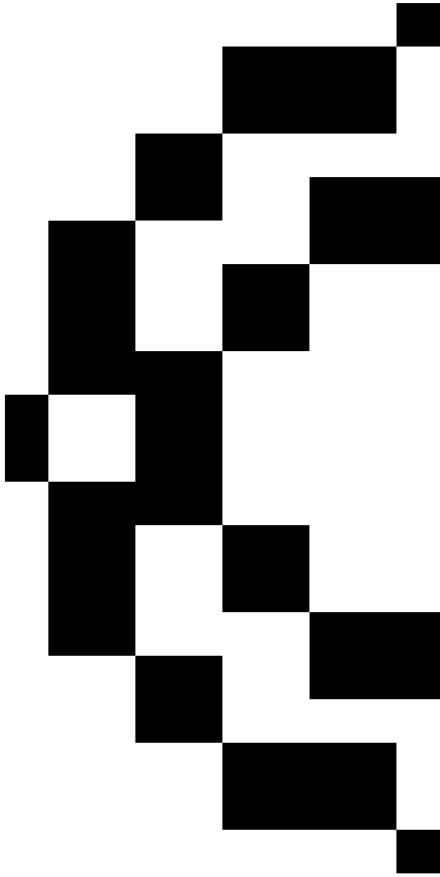
```
for i in range(1, 101):
    print(i, end=" ")
    if i % 10 == 0:
        print("That's 10 numbers printed!")
```

Exercise 5

Ask the user for a word. Check every letter in the word one by one, and each time the letter `"a"` is found, print `"Found one!"`.

Example

```
Input:  word = "banana"
Output: Found one!
        Found one!
        Found one!
```



Exercise 5 — Solution

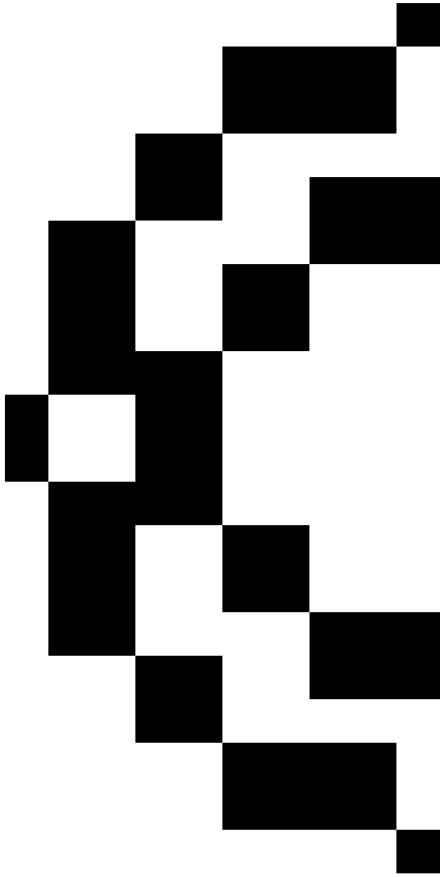
```
word = input("Insert a word: ").lower()
for letter in word:
    if letter == "a":
        print("Found one!")
```

Exercise 6

Ask the user for a word and a single letter. Print how many times that letter appears in the word.

Example

- `Input:  word = "banana", letter = "a"`
- `Output: 3`



Exercise 6 — Solution

```
word = input("Please insert a word: ")
letter = input("Please insert the letter you wish to count: ")

counter = 0
for char in word:
    if char == letter:
        counter += 1

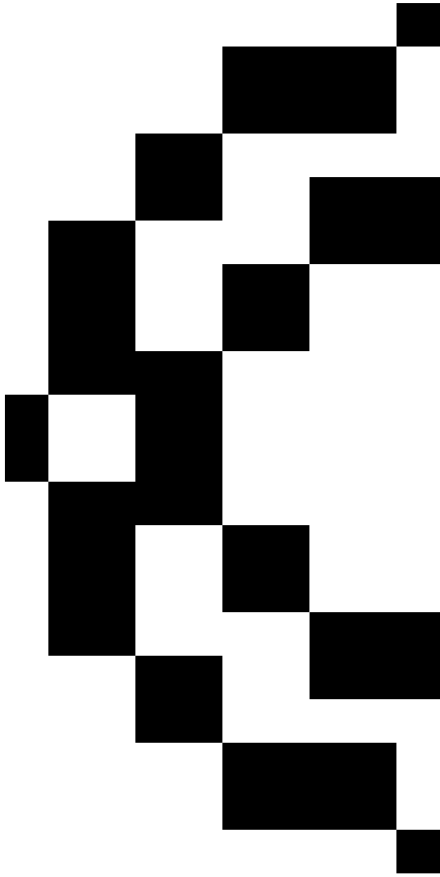
print(counter)
```

Exercise 7

Ask the user for a word. Count how many consonants (letters that are not `a`, `e`, `i`, `o`, or `u`) it contains, and print the answer.

Example

- `Input:  word = "python"`
- `Output: 4`



Exercise 7 — Solution

```
word = input("Insert a word: ")
counter = 0
for letter in word:
    if letter not in "aeiou":
        counter += 1
print(counter)
```

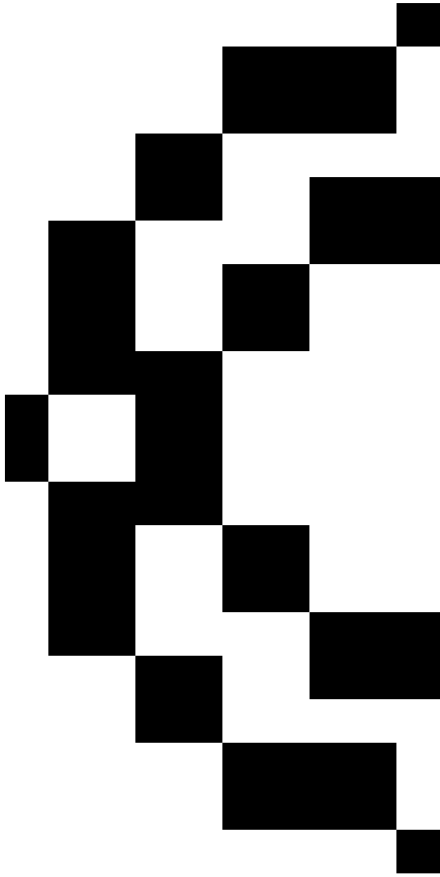
Exercise 8

Ask the user, in order: the number of classes held, the number of classes they attended, and whether they have a medical cause for absences (`"Y"` or `"N"`). A student may not sit the exam if their attendance is below 75%, unless they have a medical cause. Print their attendance percentage and whether they are allowed to sit the exam.

Examples

- `Input:  held = 20, attended = 12, medical = "Y"`
- `Output: Percentage is 60.0% with medical cause, you CAN attend the exam.`

- `Input:  held = 20, attended = 12, medical = "N"`
- `Output: Percentage is 60.0%, you CANNOT attend the exam.`



Exercise 8 — Solution

```
held = int(input("Number of classes held: "))
attended = int(input("Number of classes attended: "))
medical = input("Any medical cause? (Y or N): ")

percentage = attended / held * 100

if percentage >= 75:
    print(f"Percentage is {percentage}%, you CAN attend the exam.")
else:
    if medical == "Y" or medical == "y":
        print(f"Percentage is {percentage}% with medical cause, you CAN attend the exam.")
    else:
        print(f"Percentage is {percentage}%, you CANNOT attend the exam.")
```

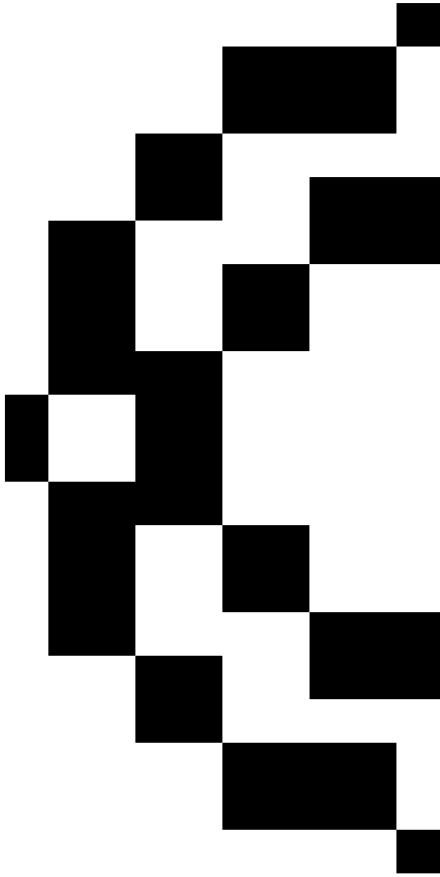
Exercise 9

Ask the user for two grades, `grade1` and `grade2`, each expected to be between 0 and 20. If either grade is outside that range, print an error message. Otherwise, ask the user to type `"Average"` or `"Greatest"`: for `"Average"`, print the average of the two grades; for `"Greatest"`, print whichever grade is larger.

Examples

- `Input:  grade1 = 14, grade2 = 17, choice = "Greatest"`
- `Output: 17.0 is the greatest`

- `Input:  grade1 = 24, grade2 = 10`
- `Output: Grade1 or Grade2 is out of the interval [0,20]`



Exercise 9 — Solution

```
grade1 = float(input("Grade1: "))
grade2 = float(input("Grade2: "))

if 0 <= grade1 <= 20 and 0 <= grade2 <= 20:
    option = input("Choose average or greatest: ").lower()
    if option == "average":
        print(f"The average is {(grade1 + grade2) / 2}")
    elif option == "greatest":
        if grade1 > grade2:
            print(f"{grade1} is the greatest")
        else:
            print(f"{grade2} is the greatest")
    else:
        print("You did not write Average or Greatest")
else:
    print("Grade1 or Grade2 is out of the interval [0,20]")
```

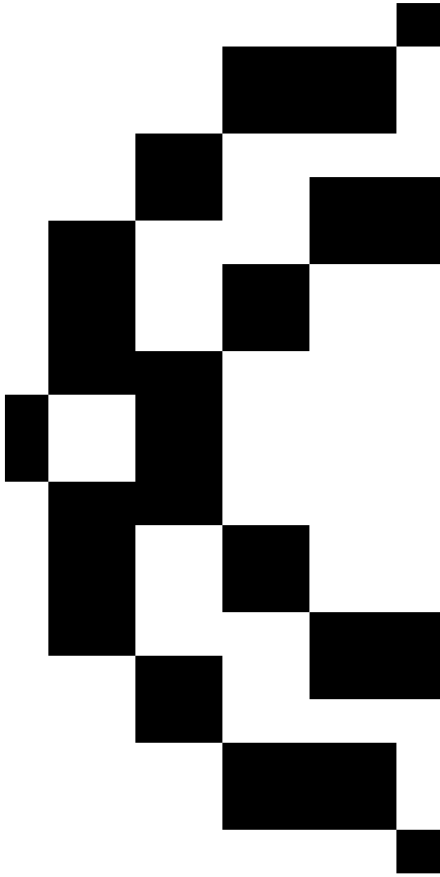
Exercise 10

A company has four departments: finance, software, management, and maintenance. Ask the user for their department and how many years they have worked there. Software and management staff get a 30% bonus with 3 or more years of experience, otherwise 25%. Finance staff get a 20% bonus with 5 or more years of experience, otherwise 15%. Maintenance staff always get a 10% bonus. Print the applicable bonus.

Examples

- `Input:  department = "software", years = 4`
- `Output: Bonus: 30%`

- `Input:  department = "finance", years = 2`
- `Output: Bonus: 15%`



Exercise 10 — Solution

```
dep = input("Department: ").lower()
years = int(input("Years: "))

if dep == "software" or dep == "management":
    if years >= 3:
        print("Bonus: 30%")
    else:
        print("Bonus: 25%")
elif dep == "finance":
    if years >= 5:
        print("Bonus: 20%")
    else:
        print("Bonus: 15%")
elif dep == "maintenance":
    print("Bonus: 10%")
else:
    print("Invalid department name")
```

NOVA IMS INFORMATION MANAGEMENT SCHOOL

Thank you

Acreditações e Certificações da NOVA IMS

Cofinanciado por